



Investigating the Reasoning Abilities of Large Language Models for Understanding Spoken Language in Interpersonal Interactions

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Abstract

This study evaluates large language models' (LLMs) reasoning capabilities in spoken language understanding (SLU) during interpersonal interactions. We incorporated several factors into LLM prompts: instructing via examples (IE), integrating domain knowledge (DK), and including context (IC). Experiments with Gemini-1.5-pro, GPT-3.5-turbo, and GPT-4o were conducted on an SLU task that classifies the degree of explanation (i.e., under-explained, succinct, comprehensive, over-explained) in job interview responses—an important step toward developing automatic interview training systems. Results demonstrate the feasibility of few-shot (1- to 4-shot) learning, with ablation studies confirming that modifications to prompts, especially when combining IE, DK, and IC, lead to further performance improvements.

Index Terms: large language models, reasoning, spoken language understanding, dialog, job interview training, degree of explanation

1. Introduction

Job interviews are the entryway to the job market, serving as a crucial step in assessing a candidate's competencies and evaluating their suitability for any specific role [1]. During interviews, interviewees must effectively broadcast their expertise through their speech content, prosody, and other verbal and non-verbal behaviors. In this regard, tracking conversations and extracting underlying meaning from candidate responses is essential for delivering targeted feedback that could enhance interview performance [2, 3].

Recent advances in LLMs have demonstrated promising reasoning abilities, such as chain-of-thought (CoT) prompting [4], that enable these models to follow the flow of dialogue, identify salient patterns, and understand semantic nuances. This involves systematically "thinking" about a problem and drawing logical conclusions based on available information [5]. In addition, the integration of LLMs in educational and professional settings is gaining traction. Recent studies indicate that generative AI tools can significantly improve workplace learning by providing personalized, real-time feedback [6]. Research has further highlighted the effectiveness of LLM-based systems in enhancing collaborative performance and streamlining communication in professional settings [7]. Moreover, conversational AI is emerging as a key component in promoting a positive workplace climate and facilitating interpersonal exchanges [8].

This paper investigates the ability of LLMs to detect ineffective communication in job interviews, an important type of spoken language communication used to assess a candidate's qualifications and communication skills. Particularly, we fo-

cus on the degree of explanation of interview responses and examine the ability of LLMs to classify a response among four classes (i.e., under-explained, succinct, comprehensive, over-explained) via incorporating several factors into the prompt, namely, instructing via examples (IE), integrating domain knowledge (DK), and integrating context (IC). Results indicate that LLMs are capable of few-shot learning for the focal task, and that the aforementioned techniques contribute to varying degrees in improving model performance. These findings underscore the potential of LLMs to process conversational data and have implications for these models ultimately providing feedback for interview training and offering insights that help users refine their responses through informed reasoning.

2. Prior Work

CoT prompting was introduced as a method to help LLMs engage in reasoning instead of simply providing answers [4]. CoT prompting was implemented with zero-shot learning by simply adding a prompt (i.e., "Let's think step by step") before each answer without including in-domain examples. Results indicate that zero-shot CoT outperformed LLMs without CoT in a variety of benchmark tasks. Further work has explored few-shot CoT by providing manually crafted reasoning of demonstrations alongside examples with step-by-step reasoning [4]. This method was effective mainly for large models with around 100 billion parameters and complex multi-step reasoning problems. However, the manual nature of few-shot CoT entails the limitation of handcrafting examples. To address this, other studies have proposed automated methods to construct demonstrations broadening the applicability of CoT prompting [9, 10].

LLMs have been recently evaluated for their ability to perform SLU tasks, which differ from written text in terms of language use, potential disfluencies, and reliance on dialog context. An initial quantitative and qualitative evaluation of ChatGPT was conducted on publicly available benchmarks for SLU tasks [11]. Similarly, ChatGPT and GPT-3.5 were found effective in zero-shot dialog understanding tasks, though they did not always match the performance of state-of-the-art fine-tuned models [12]. Results further indicated that LLMs show improved dialog understanding when multiple dialog turns are available, rather than a single one. ChatGPT and GPT-3.5 with a large number of parameters performed well on intent classification with zero- and few-shot learning, sometimes surpassing fine-tuned models on the same tasks [13]. In contrast, smaller models, such as GPT-2, OPT, and GPT-3.5 Curie, depicted lower performance in zero-shot cases. Expanding upon this, a two-stage prompting, which transforms the SLU task into a question-answering problem, achieved better results compared to LLMs without prompting [14].

Table 1: *Definitions and the number of samples (N) per class for degree of explanation.*

Class	Definition	N
Under-explained	A response that is short but does not fully answer the interviewer’s question, and often ends abruptly.	23
Succinct	A concise/short/brief/to-the-point response that fully answers the interviewers’ question.	107
Comprehensive	A detailed but to-the-point response that answers the interviewers’ questions fully and in a detailed manner.	122
Over-explained	A very long response with excess verbiage. It often goes into too much detail, which potentially affects the coherence of the answer. It can also include repetitive information.	34

The contributions of this paper are: (1) It introduces a novel method of integrating DK into LLM prompts, demonstrating the benefits of incorporating intuitive task-specific knowledge to enhance LLM performance; (2) It improves our understanding on IE and IC in LLM prompts for SLU tasks, two techniques that have been underexplored in spoken dialog settings; and (3) It expands upon existing SLU tasks by examining the LLMs’ ability to detect the degree of explanation in job interview responses, a task heavily reliant on dialog context and the structural quality of interview responses. Although our task is domain-specific, our contribution lies in showing that minimal in-context prompt augmentation enables LLMs to adapt to specialized, context-dependent tasks with few-shot learning.

3. Data Description

Our data comes from the from the VetTrain dataset [15] and include 38 military veterans who participated in a mock job interview conducted by experienced interviewers via Zoom [16]. Audio recordings and transcripts of the interviews were obtained and cleaned. Collectively, the participants responded to 286 interview questions. In order to label the degree of explanation of these responses, three annotators were employed. Based on their annotation, each response was labeled as one of four possible categories—Under-explained, Succinct, Comprehensive, and Over-explained (Table 1) [17]. The disagreement among the annotators was resolved through multiple rounds of adjudication and discussion. For each response, the final label was obtained through majority voting. The annotation process resulted in a moderate inter-annotator agreement, measured in terms of Krippendorff’s $\alpha = 0.677$.

4. Methodology

The responses belonging to the Under-explained and Succinct classes are significantly shorter in length compared to responses from the Comprehensive and Over-explained classes. In terms of total word count, Under-explained and Succinct responses contain significantly fewer words than Comprehen-

Under-explained responses have more:

* Words with political content (e.g., words and phrases related to politics, governance, and regulations).
Succinct responses have more:

* Polite expressions (e.g., words and phrases that show courtesy or deference such as "please," "thank you," and "excuse me").

* Tentative words (e.g., words indicating uncertainty or possibility such as "if", "maybe," "perhaps," and "probably").

Figure 1: *An example of text included in the prompts to integrate domain knowledge (DK) for Task 1.*

sive and Over-explained responses ($t(216.52) = -14.07, p < 0.01$). Similarly, the duration (time in seconds) of Under-explained and Succinct responses is significantly lower than that of Comprehensive and Over-explained responses ($t(214.04) = -13.70, p < 0.01$). As such, we focus on two binary classification problems. The first problem (Task 1) refers to the classification between Under-explained and Succinct classes, while the second one (Task 2) is the classification between Comprehensive and Over-explained classes.

4.1. Eliciting reasoning in LLMs

4.1.1. Instructing via examples (IE)

LLM instruction offers specific guidance to the model, allowing users to influence how the model interprets a request [4]. Here, we implemented instruction via providing reasoning steps corresponding to the degree of explanation for five randomly selected examples per class. For n -shot learning, a random subset of n examples per class was selected once and consistently used in the LLM prompts across all experiments. To avoid data leakage, we ensured that examples used in the prompt were excluded from the evaluation set; that is, performance was computed only on instances not shown to the model during prompting.

4.1.2. Integrating domain knowledge (DK)

We integrated domain knowledge to help mitigate issues of weak factuality and hallucinations often occurring in LLMs [18]. We leveraged the psycholinguistic characterization of the degree of explanation in the responses by extracting several psycholinguistic properties using the Linguistic Inquiry and Word Count (LIWC) toolbox [19]. These properties included among others general linguistic descriptors (e.g., word count), summary variables (e.g., analytical thinking), standard linguistic dimensions (e.g., pronouns), and psychological constructs (e.g., cognition), resulting in a total of 93 LIWC descriptors. We assessed the ability of each descriptor to differentiate between the two types of responses in each task via a t-test. A total of three descriptors (i.e., % words related to politics, tentativeness, politeness) were significantly different between Under-explained and Succinct responses (Task 1), while 12 descriptors (i.e., word count, prepositions, % words related to numbers, achievement, causation, negations, tentativeness, social processes, auditory processes, work, past) were significantly different between Comprehensive and Over-explained responses. These descriptors were converted into simple self-explanatory text and were added to the prompt (Fig. 1).

4.1.3. Integrating Context (IC)

We hypothesized that providing the LLM with additional context about the flow of conversation in an interview might enhance its performance. Thus, we integrated context via adding to the LLM prompt k question/response pairs from the interview that were previously assessed by the LLM and the LLMs output for each pair. Context was maintained separately for each participant and grew incrementally as the interview progresses. For example, when evaluating the n^{th} question-answer pair of an interview, the LLMs were also provided with the $(n-1)^{th}$, $(n-2)^{th}$, $\dots$, $(n-k)^{th}$ previous question/response pairs of the interview and the LLM output for each pair.

4.2. Experimental settings

We conducted experiments with the Gemini-1.5-pro, GPT-3.5-turbo, and GPT-4o in a n -shot learning framework ($n = 0, 1, \dots, 5$). All prompts included an introductory sentence setting up the context for the LLM (i.e., “*You are a text classifier. Your task is to classify the following interview response into one of two categories:*”) and the definition of the classes included in the corresponding task (i.e., Task 1 or Task 2; Table 1). Depending on the experimental setting, the prompt also included text associated with the IE (Section 4.1.1), DK (Section 4.1.2), and IC (Section 4.1.3). We also conducted experiments using pairwise combinations of the above settings, and finally incorporating all these settings in the prompt. Each query input consisted of a question-answer pair for a participant. We computed the macro-averaged F1-score that weighs each class equally regardless of its frequency. While both the tasks exhibit class imbalance, we note that our experimental setup does not involve model training and thus is not subject to learned biases in the traditional sense.

4.3. Ablation studies

We conducted three types of ablation studies: (1) We analyzed how incorporating various types of DK in the prompt influences the GPT-3.5-turbo performance. For Task 1, we included in the prompt text that corresponds to each of the three LIWC descriptors (Section 4.1.2) separately and all their combinations. For Task 2, we initially included the word count and % words related to numbers, followed by adding one at a time each of the remaining LIWC descriptors (Section 4.1.2); (2) We carried out experiments with GPT-3.5-turbo and Gemini-1.5-pro using prompts with different context lengths ($k = 1, \dots, 5$; Section 4.1.3); and (3) We experimented with three types of ordering the examples in the prompt provided to Gemini-1.5-pro to determine which would yield the best classification results: (a) all examples from the majority label (i.e., Succinct, Comprehensive) mentioned first, followed by all examples from the minority label (i.e., Under-explained, Over-explained) (M1); (2) all examples from the minority label mentioned first, followed by all examples from the majority label (M2); (3) alternating examples from each label within the prompt (M3).

5. Results

Results indicate that 1- and 2-shot learning yield good performance for both Tasks 1 and 2, while 3- and 4-shot learning also works well for Task 2 (Table 2). When examining each considered approach of LLM reasoning in isolation, IC yields better performance in the majority of cases compared to IE and DK. Combining all three methods yields further improvement for

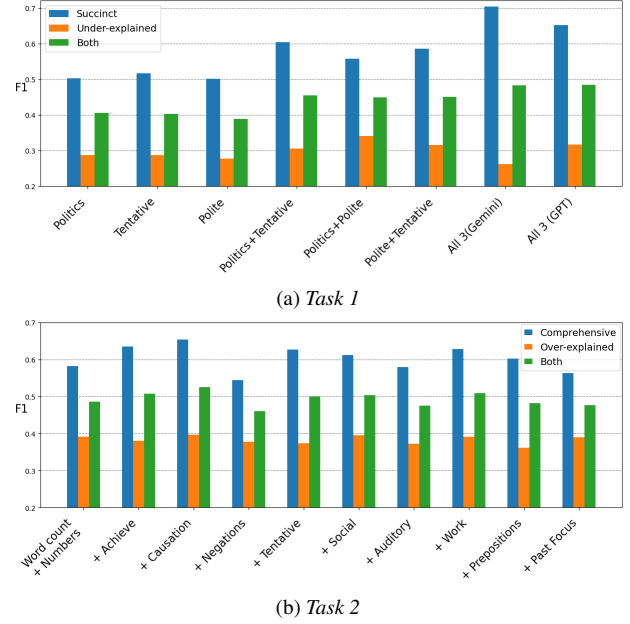


Figure 2: Macro F1-score when including different types of domain knowledge (DK) in the prompt.

Task 1 using GPT-3.5-turbo (i.e., 5% relative change in macro F1-score compared to IC alone) and Task 2 with GPT-4o (i.e., 3% relative change in macro F1-score compared to IC alone). Gemini-1.5-pro with IE+IC yields the best average performance for Task 1 (i.e., 0.50 F1-score), while GPT-3.5-turbo with IC results in the best performance for Task 2 (i.e., 0.55 F1-score). When applying IE+IC, Gemini-1.5-pro achieves the best results for both Task 1 and 2, potentially underscoring the model’s ability to handle long prompts. GPT-4o yields on average better results for Task 1 compared to GPT-3.5-turbo, while the opposite holds for Task 2. A potential reason for this might be that Task 1 is overall more challenging compared to Task 2, due to its subtlety and the overall short length of responses involved in this task, thus, it might require more sophisticated models such as GPT-4o and the recently introduced reasoning models.

In the ablation studies, integrating DK via including all three psycholinguistic descriptors for Task 1 (i.e., % words related to politics, tentativeness, and politeness) yields the best result (Fig. 2a). For Task 2, the highest macro F1-scores were achieved by combining the word count and % words related to numbers with % words related to achievement, causation, social processes, and work (Fig. 2b). This is likely because comprehensive and over-explained responses depict meaningful differences in these areas. While performance does not vary greatly based on the context length (Table 3), adding some context in the prompt ($k > 2$) generally improve the results, especially for Gemini-1.5-pro, which tends to utilize context better compared to GPT-3.5-turbo. Finally, the ordering of examples impacts the macro F1-score in Task 1 with better results when the Under-explained examples were provided before the Succinct ones in the prompt (M2) (Table 5), suggesting the importance of presenting examples from highly under-represented classes first.

6. Discussion

Findings from this study indicate that LLMs can be used to conduct few-shot learning in context-rich SLU tasks, such as assessing the degree of explanation in job interview responses.

Table 2: Macro F1-Score for classifying between Under-explained and Succinct (Task 1) and Comprehensive and Over-explained (Task 2) responses using instructing via examples (IE), domain knowledge (DK), and context (IC) from previous turns. Bold font indicates the highest average over a category (the simplest model is considered in case of a tie).

Task 1: Succinct vs Under-explained									
Model	n-shot	IE	DK	IC	IE+DK	DK+IC	IE+IC	IE+DK+IC	Mean $\pm$ Std Dev
GPT-3.5-turbo	Zero-shot	0.23	0.20	0.35	0.18	0.36	0.35	0.35	0.29 $\pm$ 0.08
	1-Shot	0.30	0.27	0.37	0.28	0.35	0.30	0.40	0.32 $\pm$ 0.05
	2-Shot	0.26	0.28	0.28	0.30	0.33	0.28	0.39	0.30 $\pm$ 0.04
	3-Shot	0.22	0.26	0.29	0.25	0.37	0.35	0.36	0.30 $\pm$ 0.06
	4-Shot	0.27	0.35	0.29	0.28	0.33	0.26	0.33	0.30 $\pm$ 0.03
	Mean $\pm$ std	0.26 $\pm$ 0.03	0.27 $\pm$ 0.05	0.32 $\pm$ 0.04	0.26 $\pm$ 0.05	0.35 $\pm$ 0.02	0.31 $\pm$ 0.04	0.37 $\pm$ 0.03	
Gemini-1.5-pro	Zero-shot	0.37	0.32	0.40	0.28	0.41	0.39	0.39	0.37 $\pm$ 0.04
	1-Shot	0.40	0.41	0.54	0.41	0.45	0.53	0.48	0.46 $\pm$ 0.05
	2-Shot	0.47	0.44	0.50	0.49	0.46	0.52	0.48	0.48 $\pm$ 0.02
	3-Shot	0.39	0.40	0.49	0.36	0.44	0.52	0.44	0.43 $\pm$ 0.05
	4-Shot	0.37	0.44	0.47	0.48	0.51	0.55	0.49	0.47 $\pm$ 0.05
	Mean $\pm$ std	0.40 $\pm$ 0.03	0.40 $\pm$ 0.04	0.48 $\pm$ 0.04	0.40 $\pm$ 0.07	0.45 $\pm$ 0.03	0.50 $\pm$ 0.05	0.46 $\pm$ 0.03	
GPT-4o	Zero-shot	0.40	0.27	0.36	0.29	0.32	0.41	0.33	0.34 $\pm$ 0.05
	1-Shot	0.43	0.36	0.41	0.34	0.35	0.42	0.36	0.38 $\pm$ 0.03
	2-Shot	0.35	0.36	0.40	0.43	0.36	0.42	0.36	0.38 $\pm$ 0.03
	3-Shot	0.39	0.33	0.38	0.35	0.34	0.37	0.37	0.36 $\pm$ 0.02
	4-Shot	0.41	0.37	0.37	0.34	0.35	0.36	0.33	0.36 $\pm$ 0.02
	Mean $\pm$ std	0.39 $\pm$ 0.02	0.33 $\pm$ 0.04	0.38 $\pm$ 0.02	0.35 $\pm$ 0.05	0.34 $\pm$ 0.01	0.39 $\pm$ 0.02	0.35 $\pm$ 0.01	
Task 2: Comprehensive vs Over-explained									
GPT-3.5-turbo	Zero-shot	0.31	0.32	0.54	0.33	0.52	0.56	0.52	0.44 $\pm$ 0.12
	1-Shot	0.23	0.40	0.59	0.31	0.55	0.54	0.54	0.45 $\pm$ 0.14
	2-Shot	0.32	0.28	0.55	0.39	0.51	0.55	0.61	0.46 $\pm$ 0.13
	3-Shot	0.28	0.37	0.53	0.41	0.51	0.49	0.52	0.44 $\pm$ 0.09
	4-Shot	0.42	0.39	0.55	0.45	0.56	0.51	0.55	0.49 $\pm$ 0.07
	Mean $\pm$ std	0.31 $\pm$ 0.07	0.35 $\pm$ 0.05	0.55 $\pm$ 0.02	0.38 $\pm$ 0.06	0.53 $\pm$ 0.02	0.53 $\pm$ 0.03	0.55 $\pm$ 0.04	
Gemini-1.5-pro	Zero-shot	0.32	0.20	0.28	0.22	0.29	0.37	0.29	0.28 $\pm$ 0.05
	1-Shot	0.28	0.27	0.28	0.36	0.35	0.44	0.32	0.33 $\pm$ 0.05
	2-Shot	0.36	0.32	0.35	0.34	0.35	0.40	0.29	0.34 $\pm$ 0.03
	3-Shot	0.43	0.35	0.40	0.46	0.36	0.44	0.37	0.40 $\pm$ 0.03
	4-Shot	0.44	0.46	0.42	0.41	0.37	0.38	0.32	0.40 $\pm$ 0.04
	Mean $\pm$ std	0.37 $\pm$ 0.06	0.32 $\pm$ 0.08	0.35 $\pm$ 0.05	0.36 $\pm$ 0.08	0.34 $\pm$ 0.02	0.41 $\pm$ 0.02	0.32 $\pm$ 0.02	
GPT-4o	Zero-shot	0.33	0.38	0.45	0.32	0.46	0.48	0.47	0.41 $\pm$ 0.06
	1-Shot	0.27	0.30	0.43	0.27	0.49	0.42	0.45	0.37 $\pm$ 0.09
	2-Shot	0.25	0.31	0.46	0.29	0.43	0.46	0.48	0.38 $\pm$ 0.09
	3-Shot	0.29	0.28	0.41	0.32	0.44	0.45	0.44	0.37 $\pm$ 0.07
	4-Shot	0.35	0.31	0.40	0.29	0.36	0.46	0.46	0.37 $\pm$ 0.06
	Mean $\pm$ std	0.29 $\pm$ 0.04	0.31 $\pm$ 0.03	0.43 $\pm$ 0.02	0.29 $\pm$ 0.02	0.43 $\pm$ 0.04	0.45 $\pm$ 0.02	0.46 $\pm$ 0.01	

Table 3: Macro F1-score with increasing context length k .

Model	Task	Context length				
		$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$
GPT-3.5Turbo	1	0.34	0.37	0.35	0.34	0.37
	2	0.51	0.53	0.52	0.53	0.55
Gemini-1.5-pro	1	0.42	0.47	0.49	0.48	0.48
	2	0.34	0.34	0.36	0.36	0.35

Table 4: Macro F1-score averaged across n -shots ($n = 1, \dots, 4$) when using three different types of ordering examples in the prompt (M1, M2, M3).

	M1	M2	M3
Task 1	0.37	0.42	0.39
Task 2	0.38	0.35	0.37

The effective augmentation of LLM prompts via IC, IE, and DK further underscores the adaptability of LLMs that can be potentially adjusted to various domains. This could be particularly useful in providing personalized feedback to job candidates during interview training, where LLMs might be able to provide personalized feedback based on a very small number of examples from a given user. This study, however, presents several limitations. First, broader generalizability remains a challenge, as the dataset used focuses on military veterans, which may not fully capture the diversity of interview styles, job sectors, or languages. Second, the study relies heavily on the construction and refinement of LLM prompts, while it would be important to explore LLM fine-tuning approaches as part of future work. Finally, the study did not investigate the explanations provided by the LLMs for their answers, which could further augment our understanding of the inner workings behind those models.

The findings from this study suggest that LLMs could play

a significant role in designing speech technologies for interview training, a relatable use case for investigating personalized and adaptive feedback. Given their ability to perform few-shot learning in context-rich SLU tasks, LLMs could assess key aspects of interview responses, such as the degree of explanation, and offer constructive suggestions and actionable feedback for improvement. The adaptability of LLM prompts, enhanced through the IC, IE, and DK techniques, indicates that these models could be tailored to different industries and job roles, making them a versatile tool for interview preparation. Additionally, the capability of LLMs to generate feedback based on minimal user examples presents an opportunity for more efficient and scalable interview coaching.

7. Conclusions

This study demonstrated the potential of few-shot learning with LLMs for conducting SLU tasks, such as classifying the degree of explanation in job interview responses. The results highlight the effectiveness of incorporating IE, DK, and IC techniques to LLMs. Future work will focus on expanding the dataset to include a broader range of interview contexts and candidate profiles, exploring additional techniques for enhancing prompt design and LLM fine-tuning, and investigating the interpretability of LLMs' decision-making processes to gain deeper insights into their reasoning.

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9. References

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